La Scuola d'Italia / Huson / Physics
27 January 2026

First & last name:
Grade:

2.4 Quiz: Motion and Kinematics

1. Round each value to three significant figures.

   (a) 0.007146 m

   (b) 52.837 kg

   (c) 6.0052 s

   (d) 18,470 N

2. Perform each operation and give the answer to 3 significant figures.

   (a) $(6.5 \times 10^3) + (8.4 \times 10^2)$

   (b) $(5.60 \times 10^{-2}) \times (3.0 \times 10^5)$

   (c) $\dfrac{9.6 \times 10^7}{4.8 \times 10^4}$

3. Convert each value. Show one line of work using unit factors.

   (a) 54.0 km/h to m/s

   (b) 12.5 m/s to km/h

4. A student walks along a straight path (forward is +). Start $x_0 = 2.5$ m.
   (a) He walks to $x = 14.5$ m. Find displacement Δx.

   (b) From $x = 14.5$ m he walks back to $x = 6.0$ m. Find Δx.

   (c) Find total displacement from start to finish.

   (d) Find total distance traveled.

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Kinematics Equations

Formulas: $v = v_0 + at$ $d = v_0 t + \frac{1}{2}at^2$ $v^2 = v_0^2 + 2ad$ $d = \frac{1}{2}(v_0 + v)t$

5. A jogger runs along a straight trail at an average velocity of 8.0 km/h for 1.5 h. How far does she travel?

6. A skateboard accelerates uniformly from rest at 1.8 $\mathrm{m/s^2}$.
   (a) What is its velocity after 5.0 s?
   (b) How far does it travel during this time?

7. A car traveling at 30 m/s applies its brakes and decelerates at 2.0 m/s^2.
   (a) How long does it take for the car to stop?

   (b) What distance does the car travel while braking?

8. A ball is thrown vertically upward with an initial velocity of 20 m/s. Use $g = 10$ m/s^2.
   (a) How long does it take to reach its maximum height?

   (b) What is the maximum height reached?

   (c) What is the ball's velocity when it returns to the thrower's hand?